

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

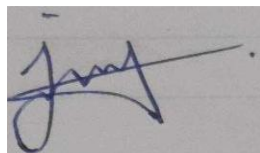
Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

- 1. Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
 - Explore containers, container architecture, benefits, and a real-world deployment example.
- 2. Serverless Computing: Reducing Infrastructure Management in the Cloud**
 - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
- 3. Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
 - Compare architectures, applications, and industry adoption scenarios.
- 4. Microservices Architecture in Cloud Environments**
 - Explain microservices design, benefits over monolithic systems, and industry implementations.
- 5. Cloud Security Fundamentals: Protecting Data and Applications**

- Cover security models, shared responsibility architecture, and real-world security practices.
- 6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
 - Analyze deployment models, business benefits, challenges, and practical examples.
- 7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
 - Discuss AI/ML cloud services, architecture, and industry applications.
- 8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**
 - Explain DevOps principles, CI/CD architecture, and practical implementation examples.
- 9. **Cloud Storage Technologies: Object, Block, and File Storage Compared**
 - Compare storage architectures, use cases, and industry adoption patterns.
- 10. **Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
 - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
- 11. **Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
 - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
- 12. **Big Data Analytics in Cloud Computing**
 - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
- 13. **Identity and Access Management (IAM) in Cloud Environments**
 - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
- 14. **Content Delivery Networks (CDNs) and Their Role in Cloud Services**
 - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.
- 15. **Server Virtualization and Its Role in Cloud Computing**

Explain virtualization architecture, hypervisors, virtual machines, resource allocation, and the benefits of virtualization in cloud data centers.

16. Cloud Storage Technologies and Architecture

Discuss cloud storage concepts, storage architectures, scalability, availability, data replication, and practical applications such as Google Drive and Amazon S3.

17. Cloud Security: Threats and Protection Mechanisms

Explain common cloud security threats, data protection, encryption, network security, security controls, and practical approaches used by cloud service providers.

18. Infrastructure as a Service (IaaS): Architecture and Applications

Explain the IaaS model, virtual computing resources, storage and networking services, advantages, limitations, and practical enterprise applications.

19. Platform as a Service (PaaS) for Cloud Application Development

Discuss PaaS architecture, development environments, application deployment, scalability, advantages, limitations, and examples of PaaS platforms.

20. Software as a Service (SaaS): Architecture and Business Applications

Explain SaaS architecture, features, advantages, subscription-based software delivery, security considerations, and examples such as Google Workspace and Microsoft 365.

Minimum slide flow:

- Title
- Aim and objective
- Core technical explanation
- Architecture or process diagram
- Methodology
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

Topic : Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems

Introduction

- Internet of Things (IoT): network of connected sensors, devices, and actuators
- Cloud computing: on-demand storage, computing, and services over the internet
- IoT + cloud: collect, transmit, process, and visualize device data
- Enables remote monitoring, automation, and data-driven decisions

Architecture focus: device, edge/gateway, network, cloud platform, application

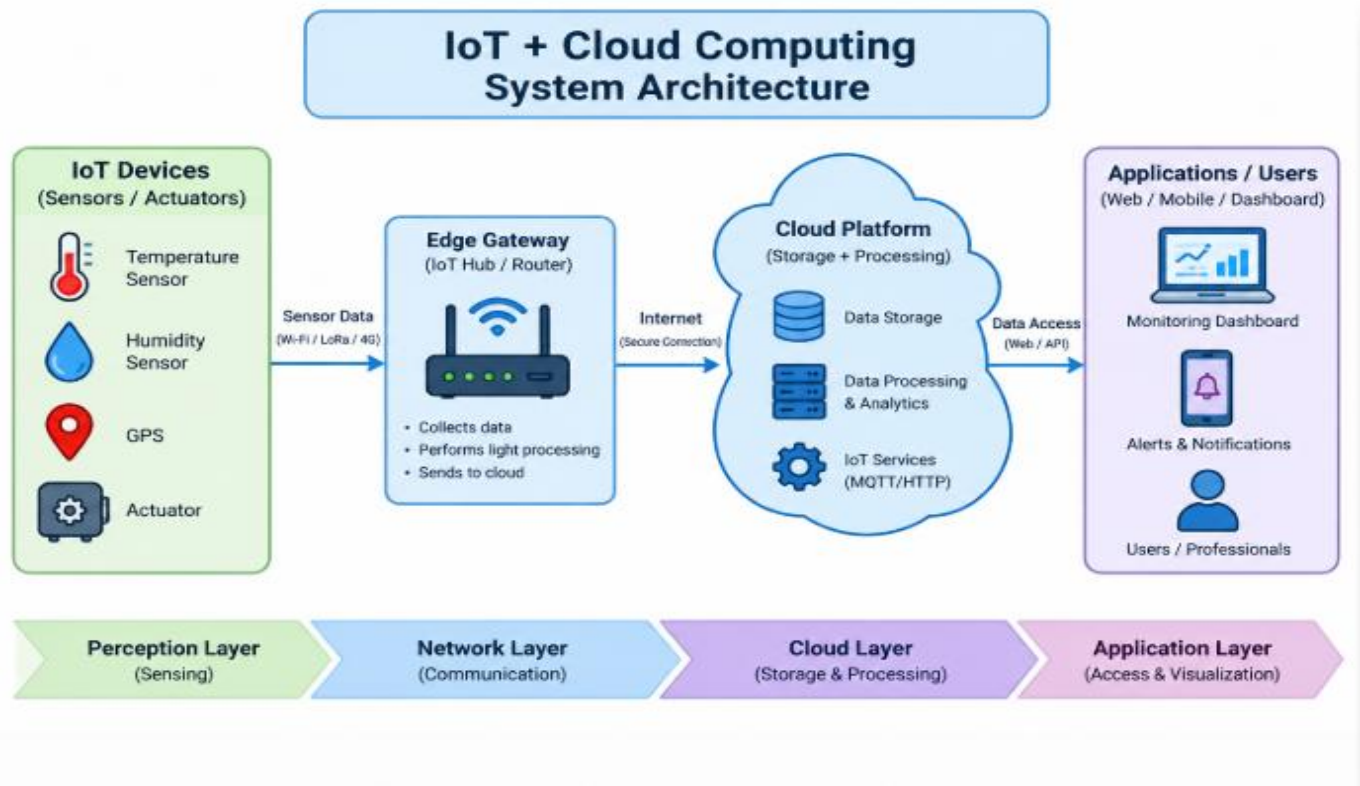
Objective

- Identify IoT and cloud architecture layers
- Explain device-to-cloud data flow
- Understand storage, processing, and analytics
- Highlight security, scalability, and real-world use

Core technical explanation:

- Large numbers of IoT devices generate continuous, heterogeneous data
- Device-only processing can be limited by compute, storage, and power
- Cloud integration introduces network dependence, latency, and security concerns
- Need: a layered architecture for reliable, scalable, secure device-to-cloud communication

Architecture or process diagram:



Real-world use case:

Smart agriculture: connected soil-moisture and weather sensors

Real-time entity: farm sensor network with cloud dashboard

- Sensors report soil moisture, temperature, and humidity
- Gateway forwards readings to a cloud IoT platform
- Cloud stores trends and compares readings with thresholds
- Dashboard supports remote irrigation monitoring
- Alerts can notify users when moisture falls below a set level

Illustrative use case; no specific deployment or measured results claimed

Conclusion and key takeaway:

IoT: Connects devices to the Internet.

Sensors: Collect data from devices.

Cloud: Stores and processes IoT data.

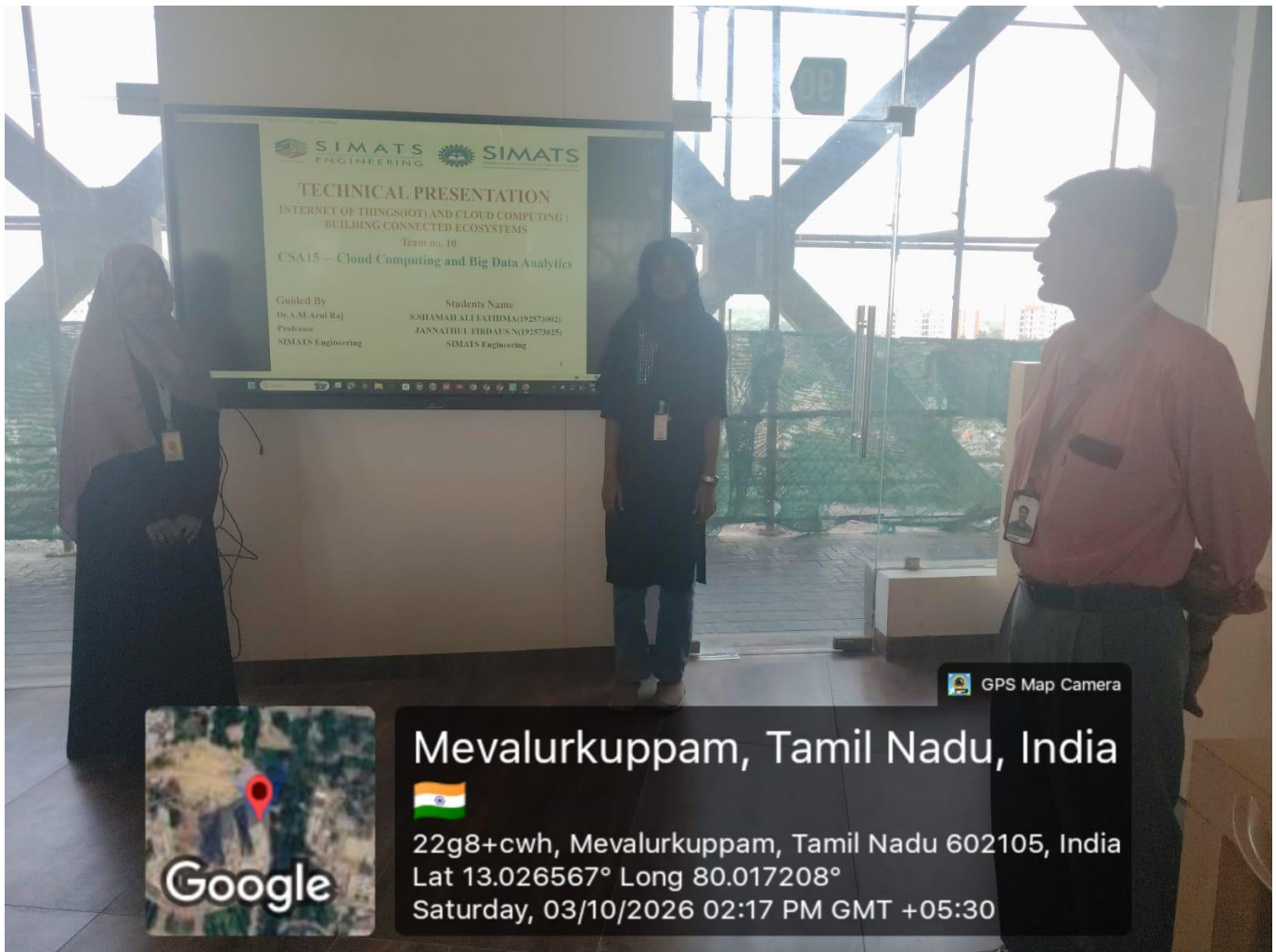
Internet: Transfers data between devices and cloud.

Benefits: Automation, real-time monitoring, scalability, and cost efficiency.

Applications: Healthcare, smart homes, industries, agriculture, and smart cities.

In short: IoT collects data → Cloud stores and processes it → Action is taken.

Photo of the presentation:



Conclusion:

IoT devices provide continuous sensing and actuation

- Cloud services add scalable storage, processing, and remote access
- Layered architecture clarifies data movement from devices to applications
- Edge processing can reduce unnecessary traffic and response delay
- Security, interoperability, and connectivity remain key design concerns
- Connected ecosystems support monitoring, automation, and informed decisions